

Edge-Adaptive Structured Pruning for Low-Latency Vision Transformers on Embedded Hardware

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Abstract—Vision Transformers (ViTs) achieve state-of-the-art accuracy across computer vision tasks, yet their quadratic self-attention complexity and high memory footprint severely constrain deployment on edge devices. Unstructured weight pruning achieves high theoretical compression but fails to yield practical hardware acceleration due to memory irregularity. In this paper, we propose EASP-ViT (Edge-Adaptive Structured Pruning for Vision Transformers), a structured pruning framework combining gradient-based attention head ranking with latency-aware channel elimination. By explicitly incorporating embedded hardware cache constraints into the pruning objective, our method produces regularized tensor geometries optimized for SIMD vector engines. Evaluated on ImageNet-1K with a Raspberry Pi 4 (ARM Cortex-A72) and an NVIDIA Jetson Nano, EASP-ViT eliminates 42.6% of model parameters and reduces FLOPs by 48.3% on ViT-Base with a marginal top-1 accuracy loss of 0.72%. Real-world latency drops by 1.84 $\times$, demonstrating that hardware-guided structured sparsity effectively closes the gap between theoretical efficiency and edge viability.

Index Terms—Vision Transformers, structured pruning, edge computing, deep learning inference, model compression, embedded AI.

I. INTRODUCTION

Deep learning architectures based on the Transformer paradigm have expanded from natural language processing to dominate modern computer vision benchmarks [1]. Despite their superior feature representation, Vision Transformers (ViTs) incur significant computational overhead. Standard configurations—such as ViT-Base, requiring approximately 86 million parameters and 17.5 GFLOPs per 224×224 frame—are ill-suited for power-constrained platforms such as autonomous drones, Internet of Things (IoT) sensors, and mobile devices.

To enable inference on edge hardware, extensive research has addressed model compression, including quantization, knowledge distillation, and weight pruning [2]. While unstructured pruning achieves sparsity exceeding 80% with negligible accuracy degradation, sparse tensor formats (e.g., Compressed Sparse Row) introduce non-deterministic memory access and control-flow divergence on SIMD and systolic-array architectures, often resulting in no wall-clock speedup [3].

Structured pruning resolves this limitation by removing coherent sub-structures, such as attention heads, linear projection dimensions, and feed-forward intermediate neurons. However,

standard structured pruning heuristics prioritize task accuracy in isolation, frequently neglecting the non-linear relationship between parameter counts and physical latency profiles on target runtimes [4], [5].

In this work, we present **EASP-ViT**, a structured pruning pipeline tailored for resource-constrained edge hardware. The primary contributions of this paper are:

- 1) A Taylor-expansion sensitivity formulation that evaluates multi-head self-attention (MHSA) importance alongside feed-forward network (FFN) channels simultaneously.
- 2) A hardware-in-the-loop latency surrogate model that penalizes configurations causing cache thrashing on target ARM and GPU architectures.
- 3) Extensive empirical validation on ImageNet-1K demonstrating a 1.84 $\times$ inference speedup on embedded hardware with negligible degradation in classification accuracy.

II. RELATED WORK

A. Vision Transformers and Computational Bottlenecks

The core building block of a ViT is the Multi-Head Self-Attention (MHSA) mechanism [1]. For an input sequence $X \in \mathbb{R}^{N \times d}$, where N denotes the number of patch tokens and d represents embedding dimensionality, the self-attention operation is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (1)$$

where Q, K, V denote queries, keys, and values, respectively. Computing the product QK^T requires $\mathcal{O}(N^2d)$ operations, which, coupled with consecutive Multi-Layer Perceptrons (MLPs), constitutes the dominant computational cost.

B. Structured Model Compression

Structured pruning methods for convolutional neural networks (CNNs) remove entire filters or residual blocks [4]. Adapting these techniques to Transformers introduces architectural challenges: pruning individual tokens alters downstream sequence lengths, while eliminating attention heads impacts the latent subspace representations. Group-sparsity regularizers and channel-pruning strategies, such as structured sparsity

learning [3] and scientific control pruning (SCOP) [5], improve computational efficiency over classic unconstrained three-stage pipelines [2]. However, prior approaches, such as patch slimming and head pruning via ℓ_1 -norm heuristics, fail to account for hardware alignment boundaries (e.g., 64-byte cache line alignments), forfeiting substantial runtime gains.

III. METHODOLOGY

Our framework optimizes the global network architecture through a three-stage procedure: sensitivity estimation, hardware-constrained optimization, and iterative fine-tuning.

A. Joint Sensitivity Estimation

Let $\mathcal{W} = \{W_h^{(l)}, W_{\text{mlp}}^{(l)}\}_{l=1}^L$ denote the set of attention head weights and MLP intermediate projection weights across all L layers. Building upon parameter sensitivity concepts in deep network pruning [2], [4], we estimate the importance score S_i of parameter block i via a first-order Taylor expansion of the loss function $\mathcal{L}(\mathcal{D})$ over validation subset $\mathcal{D}$:

$$S_i = \left| \frac{\partial \mathcal{L}(\mathcal{D})}{\partial \theta_i} \cdot \theta_i \right| \quad (2)$$

where θ_i represents the mask variable corresponding to the i -th functional unit (an individual attention head or an MLP neuron).

B. Latency-Guided Optimization

Rather than applying a uniform threshold across layers, we formulate pruning as a constrained optimization problem. Let $\tau(m)$ represent the empirical latency of sub-network mask m measured via an offline lookup table on the target deployment platform:

$$\min_m \mathcal{L}(\mathcal{W} \odot m; \mathcal{D}_{\text{train}}) \quad \text{s.t.} \quad \tau(m) \leq \mathcal{T}_{\text{target}} \quad (3)$$

where $\mathcal{T}_{\text{target}}$ denotes the latency budget, and $\odot$ represents element-wise masking. In contrast to purely statistical sparsity selection [5], units with the lowest $S_i/\Delta\tau_i$ ratios are iteratively removed until the target constraint is satisfied.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

Dataset and Baselines: We benchmark our method on ImageNet-1K using ViT-Base (16×16 patch size) as the primary baseline [1]. Comparisons are conducted against Uniform Pruning, SCOP [5], and SVD-based low-rank factorization.

Target Platforms: Evaluations are conducted on two embedded platforms:

- **Raspberry Pi 4B:** Quad-core ARM Cortex-A72 @ 1.5 GHz, 4 GB LPDDR4, running PyTorch v1.12 with ONNX Runtime (CPU backend).
- **NVIDIA Jetson Nano:** 128-core Maxwell GPU @ 921 MHz, 4 GB 64-bit LPDDR4, evaluated via TensorRT FP16 execution.

TABLE I
COMPARISON OF PRUNING APPROACHES ON ViT-BASE (IMAGENET-1K)

Method	Params (M)	GFLOPs	Top-1 (%)	RPi 4 (ms)
Unpruned ViT-B [1]	86.6	17.5	81.80	1420
Uniform Pruning	52.3	10.4	78.40	980
SCOP [5]	50.1	9.8	80.12	915
Low-Rank SVD	54.0	11.2	79.85	1030
EASP-ViT (Ours)	49.7	9.0	81.08	772

B. Quantitative Performance Comparison

Table I summarizes parameter efficiency, computation counts, top-1 validation accuracy, and physical edge latencies across the tested configurations.

As shown in Table I, EASP-ViT retains 81.08% top-1 accuracy on ImageNet-1K while dropping the per-image inference time on the Raspberry Pi 4 from 1420 ms to 772 ms—a 1.84× speedup. While traditional structured regularizers [3] and pruning pipelines [4], [5] significantly reduce theoretical FLOP counts, our hardware-aware sensitivity allocation prevents bottlenecking on critical layers where memory transfers bound execution.

C. Ablation on Attention Heads vs. FFN Layers

Our ablation shows that attention heads can be pruned unevenly across network depth. Early layers preserve cross-patch spatial context and require at least 8 active heads to prevent representation collapse. Conversely, deeper layers exhibit head redundancy, allowing up to 60% of heads in layers 9–12 to be removed with negligible degradation (< 0.2% top-1 drop).

V. CONCLUSION

In this paper, we presented EASP-ViT, an edge-adaptive structured pruning framework that bridges the gap between theoretical sparsity and empirical latency in Vision Transformers. By coupling Taylor-expansion parameter scoring with a hardware-in-the-loop latency surrogate, our framework identifies and eliminates redundant attention heads and MLP dimensions while respecting memory alignment constraints. Experimental results on embedded platforms demonstrate a 1.84× latency reduction on ImageNet-1K with only a 0.72% top-1 accuracy drop. Future research will focus on integrating 4-bit integer quantization alongside structured channel pruning for ultra-low-power microcontrollers.

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